

# [DC] User exploratory learning in a Virtual Reality museum

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## ABSTRACT

Museums are actively digitizing collections to store, distribute and share them with more audiences. The adoption of interactive technologies is promoting the spread of culture by developing creative narratives to support education and entertainment. Virtual Reality (VR) museums is an extension of physical museums that presents content in digital forms, such as photographs of museum archives. Current extended reality technologies afford the creation of interactive cultural heritage learning experience with 3D assets. This project aims to obtain an in-depth understanding of users' experiential learning in VR museums. We will investigate appropriate ways to engage users in learning culture heritage and creating personalized experiences. This is an interdisciplinary project that sits at the intersection of human-computer interaction (HCI), Museums and education. The research will provide insights and guidelines to the design and development of VR museums for effective experiential learning.

**Index Terms:** Human-centered computing [Human computer interaction (HCI)]; Interaction paradigms—Virtual reality; Human-centered computing [Human computer interaction (HCI)]; HCI design and evaluation methods—User studies

## 1 SUMMARY

With the rapid development of digital technologies, digital presentations of museums based on Virtual Reality (VR) is becoming an emerging way to provide the public immersive museum environments and interactive learning experiences [9]. Although audiences cannot interact with real cultural relics in physical museums, curators have begun to use various interactive media to increase the onsite tourists' motivation, interaction, engagement and effective learning of cultural heritage, VR museums still rarely take advantage of VR itself. Most Virtual Reality (VR) museum experiences are merely digital presentations and lack interactivity within the physical and social contexts. For instance, the physical characteristics of cultural artifacts are not afforded in the interaction design, and the social interactions with other visitors are hardly supported. These are nevertheless significant for users' motivation, interaction, engagement and effective learning of cultural heritage.

An emerging field of research in both human-computer interaction (HCI) and museums is supporting users' personalized experience. Curator enhances the value of digital assets through storytelling in exhibition curation. While artifact stories are often told by domain experts in physical museums, it is possible to engage users as 'design partners' [2] in the creation of VR museum experience, and for them to curate their own experience in VR museums. However, there are still gaps in knowledge and technology that undermine the use of user-generated content in virtual curation.

## 2 OBJECTS AND RESEARCH QUESTIONS

### 2.1 Objects

In this project, we aim to study the experiential learning in VR museums. Specifically, we have three objectives:

**O1:** Identify effective aspects of a multimedia guide that motivate user exploration in a VR museum.

**O2:** Understand the artifact interactions that contribute to users' learning in a VR museum.

**O3:** Support users' creation of personalized experience through virtual curation.

### 2.2 Research Questions

Museum guide systems should support visitors' free-choice learning in museums [4]. In VR museums, multimedia guides afford a richer array of information and interactions. It poses challenges in the effective use of multimedia to motivate users in exploring VR museums. Meanwhile, museum learning is a contextualized social activity. Personal, physical and sociocultural contexts have significant influences on museum experience [3], and should be considered in the design of multimedia guide. Therefore, we ask **RQ1:** How can a multimedia guide motivate users to explore a VR museum?

One distinct advantage of VR museum is its digital affordance of artifact interaction with vivid visualization, rich presentation and direct manipulation. However, this also creates challenges for the information display. While physical museum presents artifact information using labels with text descriptions, simply presenting textual information in a VR museum would compromise the 3D user experience. The display of artifact information in a VR museum needs to take advantage of VR's immersion, interaction and imagination features [13], using users' environmental perception, embodied cognition and the system's physical feedback to support the VR museum learning experience. Thus, we ask **RQ2:** What are the artifact interactions that contribute to users' learning in a VR museum?

Museum curation is the process of selecting, organizing and presenting museum artifacts for effective storytelling, which helps create an emotional bond between the museum and the public. Unlike the limited spaces of a physical museum, a VR museum has an infinite virtual space. As a result, there is more room for museum curators to design and host exhibitions, and for users to create and share their personal experiences. User engagement in virtual curation indicates active experimentation [8] and the sharing of personal meaning-making [13], which are also an important parts of experiential learning. This brings us to **RQ3:** How to support virtual curation and users' personalized experiences in a VR museum?

## 3 THEORETICAL FRAMEWORK AND METHODS

The project will follow the design science research paradigm [6]. We use theories in HCI and learning when designing and developing VR museum systems, which are evaluated to refine the design and development. The research will lead to applications in appropriate environments and contribute to the knowledge base in HCI and museum.

**Theories** We start with the analysis of theories that are related to experiential learning in VR museums, including the Experiential

Learning Cycle [6], Contextual Model of Learning [3], and the emotion model [5]. These will lay a solid theoretical foundation for the project.

**Task Analysis** To understand behavioral and cognitive tasks in VR museums, we conduct Hierarchical Task Analysis (HTA) [1] and Goal, Operator, Method and Selection (GOMS) task analysis [7]. This will guide the design of the multimedia guide, artifact interaction and the tools to support virtual curation.

**System design and development** We will adopt a participatory design approach [12]. Domain experts will be invited to interviews to obtain in-depth understanding of their views on museum learning, virtual and online exhibitions, curation practices, and digital transformation in museums. Potential target users and stakeholders will be involved to understand their requirements and expectations on multimedia guide and artifact interactions in a VR museum. The system will be developed in Unity with 3D models constructed and edited in Blender.

**Data collection and analysis** Two objective measures will be adopted for data collection. One is user activity monitoring. We will embed scripts in the system to record the real-time location of users and artifacts and their interactions within a VR museum. It will support the analysis of user interactions and task performance. The other is electroencephalogram (EEG) monitoring. The brain signals captured can be used to analyze mental workload and user engagement [10]. Questionnaire-based approach and interviews will also be used to measure learning outcomes, understand users' subjective feelings, and cross-examine the objective data analysis results. Quantitative data will be analyzed using statistical analysis; qualitative data will be analyzed using theme-based content analysis [11].

**Experimental study** The developed systems will be evaluated in experimental studies. In the first study, we aim to answer **RQ1** and achieve **O1**. Participants will be invited to experience two different types of multimedia guides: one that simulates current museum guides and the other being adapted for the use in a VR museum. A between-subjects study will be conducted to compare the two guide systems and participant motivation and learning outcomes will be analyzed. In the second study, we aim to answer **RQ2** and achieve **O2**, focusing on users' interaction with virtual artifacts. Artifact interaction data captured in user activity monitoring and subjective evaluations will be analyzed and compared during several design iterations addressing the task analysis results. Within-subjects studies will be conducted to measure perceived workload and task performance. For multiuser scenarios, social presence and collaboration performance will be measured. In the third study, we aim to answer **RQ3** and achieve **O3**. User activity data collected in the previous studies will contribute to the formulation of user interaction model. We incorporate the lessons learned to develop and evaluate the tools and interaction techniques to support virtual curation in a VR museum. For this part, we focus on the evaluation of virtual curation practices, and study how it contributes to users' personalized experience using interviews and focus groups.

#### 4 CONTRIBUTION OF THE PROPOSED RESEARCH

**Theoretical contribution** Our interdisciplinary project combines the theoretical foundations and methods that are used in research on science and technology and arts and humanities. Our work will fill the knowledge gap of experiential learning in VR museums and contribute to a series of guidelines that inform the future research on VR museums.

**Empirical evidence and methodological implications** This project triangulates findings in existing theories and related work with quantitative and qualitative experimental studies. The large-scale objective data collected in this project will contribute significantly to the empirical evidence of research findings. The research methods used in this project are applicable and of reference significance to future VR studies.

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**Design implications** Following the design science research paradigm and the participatory design process, the project will have a fair amount of discussion about the design implications. This part will be of interest to researchers and practitioners in the fields of HCI, museums and education who are interested to design, develop and adopt digital technologies in related contexts. GLAMs, tourism and education. This research is of significant value for galleries, libraries, archives and museums (GLAMs) and provides insights into the presentation of digital assets in an interactive and engaging way. The use of modern technologies will attract young people to experience and learn about traditional culture and motivate users to have onsite visits to tourism sites. Our research demonstrates the use of learning theories in the intersection of VR and museums, which shows significant value and impact on VR education in general.

To summarize, the proposed research is original and of significant value in both research knowledge and applications. This project will bring insights for researchers, practitioners and educators to design, develop and adopt VR systems and support content creation within the virtual environment. It will also broaden the application of VR in the domains of GLAMs, tourism and education.

#### REFERENCES

- [1] J. Annett. Hierarchical task analysis. *Handbook of cognitive task design*, 2:17–35, 2003.
- [2] V. Cesário, D. Petrelli, and V. Nisi. Teenage visitor experience: Classification of behavioral dynamics in museums. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, pp. 1–13, 2020.
- [3] L. D. Dierking. The role of context in children's learning from objects and experiences. *Perspectives on object-centered learning in museums*, pp. 3–18, 2002.
- [4] J. Falk. Free-choice learning: An alternative to term to informal learning? *Informal learning environments research newsletter*, 2(1):2, 1998.
- [5] A. J. Gerber, J. Posner, D. Gorman, T. Colibazzi, S. Yu, Z. Wang, A. Kangarlou, H. Zhu, J. Russell, and B. S. Peterson. An affective circumplex model of neural systems subserving valence, arousal, and cognitive overlay during the appraisal of emotional faces. *Neuropsychologia*, 46(8):2129–2139, 2008.
- [6] A. R. Hevner, S. T. March, J. Park, and S. Ram. Design science in information systems research. *Management Information Systems Quarterly*, 28(1):6, 2008.
- [7] D. Kieras. Goms models for task analysis. *The handbook of task analysis for human-computer interaction*, pp. 83–116, 2004.
- [8] D. A. Kolb. *Experiential learning: Experience as the source of learning and development*. FT press, 2014.
- [9] Y. Li, T. Ma, and E. Ch'Ng. Enhancing vr experiential learning through the design of embodied interaction in a shared virtual environment. 2019.
- [10] T. McMahan, I. Parberry, and T. D. Parsons. Evaluating player task engagement and arousal using electroencephalography. *Procedia Manufacturing*, 3:2303–2310, 2015.
- [11] H. Neale and S. Nichols. Theme-based content analysis: a flexible method for virtual environment evaluation. *International journal of human-computer studies*, 55(2):167–189, 2001.
- [12] D. Schuler and A. Namioka. *Participatory design: Principles and practices*. CRC Press, 1993.
- [13] T. B. Sheridan. Interaction, imagination and immersion some research needs. In *Proceedings of the ACM symposium on Virtual reality software and technology*, pp. 1–7, 2000.